

Evaluating Limits



Direct substitution and algebraic techniques

1 Evaluate each limit:

a $\lim_{x \rightarrow 1} (3x^2 - 2x + 5)$

b $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

c $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$

2 Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x^2 - 9}$.

Trigonometric and special limits

3 Using $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, evaluate:

a $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$

b $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$

c $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x}$

4 Evaluate $\lim_{x \rightarrow \infty} \frac{5x^3 - 2x + 1}{2x^3 + 7}$.

Limits that fail to exist and the squeeze theorem

5 Explain why $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist.

6 Use the squeeze theorem to evaluate $\lim_{x \rightarrow 0} x^2 \cos\left(\frac{1}{x}\right)$.

Limits with absolute value and piecewise functions

7 Evaluate $\lim_{x \rightarrow 3} |x - 3| \cdot \frac{1}{x}$.

8 For $f(x) = 2x + 1$ if $x < 3$, and $f(x) = x^2 - 2$ if $x \geq 3$, evaluate $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$, and state whether $\lim_{x \rightarrow 3} f(x)$ exists.

More algebraic limit techniques

- 9 Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$ by factoring with the difference-of-cubes formula.
- 10 Evaluate $\lim_{h \rightarrow 0} \frac{(2+h)^2 - 4}{h}$.